Appendix: Clinical-Anchored Multi-Omics Graph Learning

Author information to be completed

This is a separate author-review supplement. It is not part of the four-page ISBI manuscript and is not assumed eligible for extra technical submission pages.

1 Data alignment and evidence boundaries

The primary cohort contains 1,416 unique primary-tumour patients with three molecular assays and valid overall survival; 830 died. Clinical records numbered 2,509, and 1,980 had usable survival. Three assay intersections contained 1,417 clinical samples before survival/primary-tumour filtering. The clinical-table exclusions therefore total 1,093 for the final complete-assay analysis. Source-file hashes and exact feature counts are in results/data_audit.json. No outcomes, recurrence fields, future treatment information or cohort labels are used as predictive clinical inputs. Cohort labels are reserved for evaluation.

The supplied TCGA/MOGONET matrices lack patient IDs and are not joined by row order to METABRIC. BUSI and BUSBRA are unrelated patients and are not joined to molecular records. CNA replaces the preliminary miRNA view. Temporal graphs, simulated visits, clinical notes and paired imaging are not evaluated.

Source group	Patients	Deaths
1.000	386.000	173.000
2.000	225.000	121.000
3.000	495.000	357.000
4.000	160.000	80.000
5.000	150.000	99.000

View	Raw features	Missing cells
mRNA	20603	10
Methylation	13188	553726
CNA	22544	557

2 All fixed parameters and search settings

Parameter	Value
outer_folds	5
validation_fraction	0.2
split_seed	42
seeds	[42, 2026, 3407]
trials	12
epochs	300
patience	30
bootstrap	1000
confidence_level	0.95
concordance_tie_tolerance	1e-08
prediction_horizons_months	[12, 36, 60]
ipcw_tau_training_quantile	0.9
minimum_censoring_survival	0.05
calibration_bins	5
risk_group_rule	"training median per fold and seed; three-seed majority vote"
feature_counts	[500, 1000]
ranks	[4, 8, 16]

ridges	[0.01, 0.1, 1.0]
anchors	[1.0, 2.0]
neighbors	[5, 10, 20]
hidden_sizes	[32, 64]
learning_rates	[0.001, 0.003, 0.01]
mask_rates	[0.0, 0.2, 0.4]
dropout	0.3
gcn_layers	2
activation	"ReLU"
layernorm_epsilon	1e-05
weight_decay	0.001
optimizer	"AdamW"
optimizer_betas	[0.9, 0.999]
optimizer_epsilon	1e-08
scheduler	"none"
gradient_clip	5.0
empty_mask_fallback	"mRNA retained"
cox_ties	"Breslow"
gcca_centering	"training mean"
gcca_projection_scaling	"training standard deviation"
graph_metric	"cosine"
graph_adjacency	"symmetric binary training kNN with self loops"
query_edges	"training to query only; frozen reference states and degrees"
feature_max_missing_fraction	0.2
feature_clip_standard_deviations	5.0
methylation_epsilon	1e-05
tfidf_min_df	2
tfidf_max_features	10000
tfidf_ngram_range	[1, 2]
tfidf_norm	"l2"
tfidf_lowercase	true
tfidf_smooth_idf	true
tfidf_sublinear_tf	false
text_svd_max_dimensions	128
cox_l1_ratio	0.1
cox_max_iterations	10000
cox_tolerance	1e-06
rsf_trees	200
rsf_jobs	4
rsf_low_memory	true
rsf_survival	"bootstrap-weighted terminal-node Kaplan-Meier; native-equivalence tested"
rsf_max_features	"sqrt"
multisurv_encoder_dimensions	[128, 32]
multisurv_intervals	30
multisurv_interval_months	12
integrao_fusing_iterations	20
integrao_alignment_epochs	300
integrao_hidden	128
integrao_embedding	8
integrao_neighbors	10
integrao_cox_tolerance	1e-07
threads	2

The twelve candidate dictionaries are in search_candidates.json; selected_parameters.csv stores the actual fold/seed settings. GCCA projection coordinates are standardized using training projection standard deviations. Clinical numerical values

are inserted into fixed field–value templates; missing values are "unknown". Structured clinical encoding uses training median filling, numerical missingness indicators, standardization and one-hot categories with unknown categories ignored. TF-IDF retains tokens containing digits and decimals; no pretrained language encoder is used.

Coxnet uses l1 ratio 0.1, at most 10,000 iterations and tolerance 10^{-6}. Its size is the number of nonzero coefficients. Neural size is the number of trainable scalar parameters. Survival-forest size is total tree-node count, which is not directly comparable to neural parameter count. The forest uses 200 trees, square-root feature sampling and four local workers. Its memory- efficient implementation reconstructs terminal-node bootstrap-weighted Kaplan–Meier probabilities only at requested horizons; agreement with native full-memory survival curves is tested. Twelve configurations are an upper budget, not an exhaustive Cartesian search. Some candidate choices irrelevant to a particular baseline are retained in the audit record but do not change that model; duplicate effective forest configurations are evaluated once. The structured clinical Cox control was added after initial fold inspection to strengthen the clinical comparison. This is an exploratory, non-preregistered study.

MultiSurv-style encoders have a 128-unit first layer and 32/64-unit output, elementwise maximum fusion and a same-width hidden prediction layer producing 30 twelve-month hazards. Its scalar C-index risk is negative summed survival across intervals, whereas time-dependent AUC below uses this same global marker. Its original pan-cancer architecture and original code are not reproduced exactly. MOGONET-style encoders use modality-specific two-layer graphs and a concatenated Cox head; no classification-label VCDN is used. DeepSurv-style denotes a two-hidden-layer ReLU/dropout MLP trained with Cox loss.

IntegrAO uses the official 0.1.3 implementation with fixed representation settings: 128 hidden units, 8 output dimensions, 10 neighbors, 20 diffusion steps, and 300 alignment epochs. Adam uses its original schedule from 0.1, decayed by ten each 100 epochs to a minimum 0.001; alignment MSE weight is one and t-SNE early exaggeration is removed at epoch 100. A Coxnet head alpha is selected from twelve log-spaced values 10^{-3} to 10^{1}; its Coxnet head uses tolerance 10^{-7} (the other Coxnet baselines use 10^{-6}). New patients receive one-way incoming edges from training neighbors only. Evaluation mode freezes batch normalization; query-batch independence is asserted. These survival and inductive adapters are explicit departures from the original downstream task. Exact package versions are in requirements.txt and environment.json.

3 Full discrimination, time-dependent metrics and model size

Configuration	C-index	CI low	CI high
Structured clinical Cox	0.656	0.634	0.675
Structured clinical input	0.655	0.635	0.672
Random survival forest	0.620	0.600	0.639
mRNA Cox	0.619	0.599	0.640
Pairwise CCA + Cox	0.619	0.599	0.639
Templated clinical Cox	0.618	0.596	0.638
Without raw residual	0.609	0.590	0.627
DeepSurv-style MLP	0.607	0.587	0.626
Shared-anchor GCCA + graph	0.607	0.587	0.625
MOGONET-style + Cox	0.605	0.585	0.625
Concatenation Cox	0.604	0.584	0.625
Without graph	0.602	0.582	0.620
Equal GCCA weights	0.601	0.581	0.619
GCCA + Cox	0.598	0.576	0.621
Without view masking	0.597	0.577	0.616
Original pairwise CCA + graph	0.591	0.575	0.609
MultiSurv-style (scaled)	0.586	0.567	0.605
Official IntegrAO + Cox adapter	0.576	0.557	0.597
Methylation Cox	0.544	0.523	0.567
CNA Cox	0.515	0.492	0.538

Configuration	IPCW C	AUC 1y	AUC 3y	AUC 5y
Structured clinical Cox	0.654	0.733	0.674	0.687
Structured clinical input	0.645	0.650	0.680	0.688
Random survival forest	0.600	0.796	0.678	0.669
mRNA Cox	0.607	0.580	0.655	0.661
Pairwise CCA + Cox	0.603	0.598	0.658	0.651
Templated clinical Cox	0.605	0.747	0.678	0.666
Without raw residual	0.594	0.601	0.668	0.655
DeepSurv-style MLP	0.597	0.650	0.633	0.645
Shared-anchor GCCA + graph	0.590	0.671	0.669	0.652
MOGONET-style + Cox	0.595	0.731	0.640	0.638
Concatenation Cox	0.592	0.741	0.638	0.631
Without graph	0.586	0.691	0.666	0.647
Equal GCCA weights	0.586	0.661	0.652	0.648
GCCA + Cox	0.581	0.679	0.665	0.639
Without view masking	0.583	0.708	0.659	0.643
Original pairwise CCA + graph	0.579	0.640	0.639	0.617
MultiSurv-style (scaled)	0.576	0.623	0.593	0.607
Official IntegrAO + Cox adapter	0.556	0.626	0.641	0.628
Methylation Cox	0.543	0.623	0.524	0.542
CNA Cox	0.512	0.451	0.547	0.531

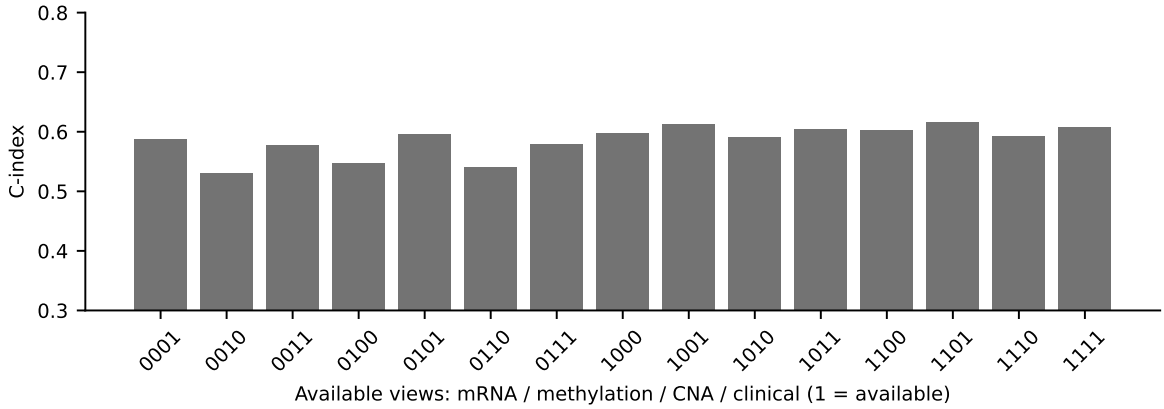
Configuration	BS 1y	BS 3y	BS 5y	Size
Structured clinical Cox	0.015	0.091	0.157	22.000
Structured clinical input	0.015	0.091	0.158	11045.000
Random survival forest	0.015	0.090	0.161	17614.000
mRNA Cox	0.015	0.096	0.165	472.000
Pairwise CCA + Cox	0.015	0.093	0.167	14.000
Templated clinical Cox	0.015	0.093	0.163	83.000
Without raw residual	0.015	0.093	0.165	11009.000
DeepSurv-style MLP	0.015	0.097	0.176	204481.000
Shared-anchor GCCA + graph	0.015	0.092	0.164	11045.000
MOGONET-style + Cox	0.015	0.094	0.171	121345.000
Concatenation Cox	0.017	0.111	0.198	465.000
Without graph	0.015	0.092	0.166	11045.000
Equal GCCA weights	0.015	0.093	0.165	11045.000
GCCA + Cox	0.015	0.092	0.165	19.000
Without view masking	0.015	0.093	0.167	11045.000
Original pairwise CCA + graph	0.015	0.096	0.180	10005.000
MultiSurv-style (scaled)	0.015	0.094	0.174	419454.000
Official IntegrAO + Cox adapter	0.015	0.094	0.169	809664.000
Methylation Cox	0.024	0.148	0.241	401.000
CNA Cox	0.017	0.106	0.188	420.000

Harrell concordance is the mean across five held-out folds and three seeds. Intervals use 1,000 patient-level bootstrap resamples separately within each held-out fold, with identical resamples for all methods. Seed-wise concordance contributions are averaged within a resample. These intervals are conditional on fitted models, not nested-bootstrap retraining intervals. Pairwise differences are two-sided and Holm-adjusted over all comparisons against the proposed configuration, including ablations. Raw test outcomes are retained for Harrell concordance. For time-dependent metrics, follow-up beyond training support is administratively censored just below the largest training follow-up. Censoring survival at each horizon must exceed 0.05; a nonestimable metric is NA. Fold-average time-dependent metrics use available estimable folds. No finite-sample selective-risk guarantee is inferred.

Comparator	Difference	CI low	CI high	Holm p
Methylation Cox	0.063	0.038	0.088	0.038
CNA Cox	0.092	0.066	0.118	0.038
Structured clinical input	-0.048	-0.065	-0.031	0.038
Structured clinical Cox	-0.049	-0.075	-0.025	0.038
Official IntegrAO + Cox adapter	0.031	0.010	0.052	0.060
Without view masking	0.009	0.003	0.015	0.060
MultiSurv-style (scaled)	0.021	0.001	0.041	0.519
Without graph	0.005	-0.000	0.011	0.839
Original pairwise CCA + graph	0.015	-0.003	0.035	1.000
Equal GCCA weights	0.006	-0.002	0.013	1.000
Random survival forest	-0.012	-0.031	0.005	1.000
Pairwise CCA + Cox	-0.012	-0.031	0.007	1.000
mRNA Cox	-0.012	-0.033	0.009	1.000
Templated clinical Cox	-0.011	-0.034	0.012	1.000
GCCA + Cox	0.008	-0.009	0.026	1.000
Without raw residual	-0.002	-0.008	0.004	1.000
Concatenation Cox	0.003	-0.019	0.025	1.000
MOGONET-style + Cox	0.002	-0.017	0.021	1.000
DeepSurv-style MLP	-0.000	-0.019	0.018	1.000

4 All missing-view combinations

mRNA/Meth/CNA/Clinical	Proposed	Pairwise graph	No masking
0001	0.587	0.562	0.578
0010	0.531	0.522	0.512
0011	0.577	0.560	0.562
0100	0.547	0.504	0.534
0101	0.596	0.549	0.584
0110	0.541	0.510	0.522
0111	0.579	0.545	0.564
1000	0.598	0.590	0.587
1001	0.613	0.597	0.608
1010	0.590	0.589	0.575
1011	0.604	0.598	0.595
1100	0.602	0.581	0.591
1101	0.617	0.591	0.609
1110	0.592	0.578	0.578
1111	0.607	0.591	0.597



Configuration	Missing probability	C-index
Without view masking	0.100	0.591
Without view masking	0.300	0.584
Without view masking	0.500	0.568
Original pairwise CCA + graph	0.100	0.587
Original pairwise CCA + graph	0.300	0.574
Original pairwise CCA + graph	0.500	0.565
Shared-anchor GCCA + graph	0.100	0.599
Shared-anchor GCCA + graph	0.300	0.599
Shared-anchor GCCA + graph	0.500	0.587

5 Source-group holdouts

5.1 cohort1

Configuration	Patients	Deaths	C-index	Seed SD
Templated clinical Cox	386	173	0.612	0.000
Structured clinical Cox	386	173	0.722	0.000
Concatenation Cox	386	173	0.620	0.000
GCCA + Cox	386	173	0.667	0.000
Official IntegrAO + Cox adapter	386	173	0.609	0.009
Original pairwise CCA + graph	386	173	0.653	0.034
Shared-anchor GCCA + graph	386	173	0.658	0.019

5.2 cohort2

Configuration	Patients	Deaths	C-index	Seed SD
Templated clinical Cox	225	121	0.612	0.000
Structured clinical Cox	225	121	0.605	0.000
Concatenation Cox	225	121	0.604	0.000
GCCA + Cox	225	121	0.611	0.000
Official IntegrAO + Cox adapter	225	121	0.584	0.008
Original pairwise CCA + graph	225	121	0.552	0.019
Shared-anchor GCCA + graph	225	121	0.607	0.016

5.3 cohort3

Configuration	Patients	Deaths	C-index	Seed SD
Templated clinical Cox	495	357	0.594	0.000
Structured clinical Cox	495	357	0.620	0.000
Concatenation Cox	495	357	0.581	0.000
GCCA + Cox	495	357	0.575	0.000
Official IntegrAO + Cox adapter	495	357	0.556	0.009
Original pairwise CCA + graph	495	357	0.567	0.011
Shared-anchor GCCA + graph	495	357	0.584	0.003

5.4 cohort4

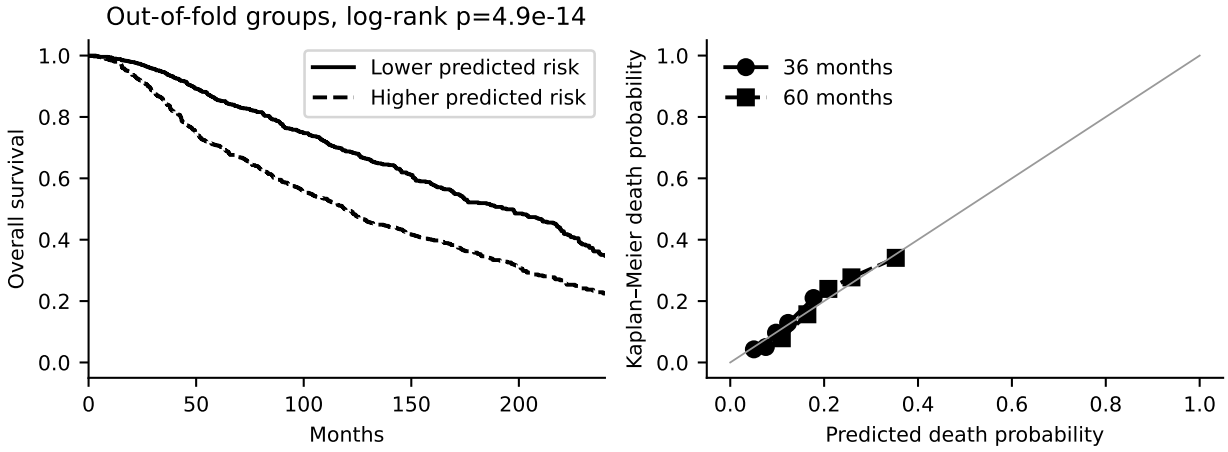
Configuration	Patients	Deaths	C-index	Seed SD
Templated clinical Cox	160	80	0.593	0.000
Structured clinical Cox	160	80	0.659	0.000
Concatenation Cox	160	80	0.651	0.000
GCCA + Cox	160	80	0.624	0.000
Official IntegrAO + Cox adapter	160	80	0.600	0.014
Original pairwise CCA + graph	160	80	0.611	0.025
Shared-anchor GCCA + graph	160	80	0.641	0.016

5.5 cohort5

Configuration	Patients	Deaths	C-index	Seed SD
Templated clinical Cox	150	99	0.558	0.000
Structured clinical Cox	150	99	0.630	0.000
Concatenation Cox	150	99	0.574	0.000
GCCA + Cox	150	99	0.540	0.000
Official IntegrAO + Cox adapter	150	99	0.526	0.010
Original pairwise CCA + graph	150	99	0.552	0.011
Shared-anchor GCCA + graph	150	99	0.562	0.014

These are leave-one-source-group-out analyses within METABRIC, not independently collected external cohorts. Their preprocessing and graph references exclude the held-out group. Validation is 20% of remaining patients, event-stratified with seed 42. The same 12-configuration upper search budget is used except for the documented fixed IntegrAO representation.

6 Calibration, risk groups and interpretability limits



Each training fold's median predicted risk defines its test high-risk group; three seed-wise decisions are combined by majority vote. Calibration averages the three seed-specific survival probabilities within each held-out fold, then compares prediction quintiles with Kaplan–Meier event probabilities. These plots are descriptive, and risk-group separation is not a replacement for discrimination or calibration metrics. GCCA loadings are linear associations and should not be described as causal biomarkers. The current study does not establish clinically validated biomarkers or graph attention: the implemented graph is a GCN, not an attention network.

7 Relationship to supplied reference PDFs

The CXR-LT reports (Holste 2024; Dong 2026 overview and full report) concern long-tailed chest-X-ray classification. They motivate careful cross-domain testing but do not supply survival comparison numbers. The Aperstein 2025 preprint and 2026 Scientific Reports version concern rejection in multi-pathology chest-X-ray classification; the later article is not counted as an independent second study. Geifman and El-Yaniv (2017) concerns selective classification. Bates et al. (2021) concerns risk-controlling prediction sets; Angelopoulos et al. (Learn Then Test, arXiv 2021) concerns finite-sample calibration through hypothesis testing. Their guarantees are not transferred to censored Cox outcomes in this implementation. No chest-X-ray experiment, selective survival guarantee, or numerical baseline result is invented.

8 Per-fold audit and reproducibility

The complete fold-results CSV contains all patient counts, event counts, metrics, training seconds, selected epoch, model size and exact configuration. All forty selected main-CV Cox solutions were re-fitted in a convergence audit. They reproduced their original C-indices to numerical precision and emitted no solver warnings. Failed numerical search candidates were not assigned performance values and were excluded from selection; the attempt log retains their status. Interrupted, uncompleted fits were rerun. The following table includes each completed main-CV fold/seed result.

Configuration	Fold	Seed	C-index	Epoch
clinical_cox	cv0	42	0.6354	0
clinical_cox	cv0	2026	0.6354	0
clinical_cox	cv0	3407	0.6354	0
clinical_cox	cv1	42	0.5896	0
clinical_cox	cv1	2026	0.5896	0
clinical_cox	cv1	3407	0.5896	0
clinical_cox	cv2	42	0.5985	0
clinical_cox	cv2	2026	0.5985	0
clinical_cox	cv2	3407	0.5985	0
clinical_cox	cv3	42	0.6290	0
clinical_cox	cv3	2026	0.6290	0

clinical_cox	cv3	3407	0.6290	0
clinical_cox	cv4	42	0.6395	0
clinical_cox	cv4	2026	0.6395	0
clinical_cox	cv4	3407	0.6395	0
clinical_structured_cox	cv0	42	0.6479	0
clinical_structured_cox	cv0	2026	0.6479	0
clinical_structured_cox	cv0	3407	0.6479	0
clinical_structured_cox	cv1	42	0.6453	0
clinical_structured_cox	cv1	2026	0.6453	0
clinical_structured_cox	cv1	3407	0.6453	0
clinical_structured_cox	cv2	42	0.6487	0
clinical_structured_cox	cv2	2026	0.6487	0
clinical_structured_cox	cv2	3407	0.6487	0
clinical_structured_cox	cv3	42	0.6772	0
clinical_structured_cox	cv3	2026	0.6772	0
clinical_structured_cox	cv3	3407	0.6772	0
clinical_structured_cox	cv4	42	0.6601	0
clinical_structured_cox	cv4	2026	0.6601	0
clinical_structured_cox	cv4	3407	0.6601	0
cna_cox	cv0	42	0.4735	0
cna_cox	cv0	2026	0.4735	0
cna_cox	cv0	3407	0.4735	0
cna_cox	cv1	42	0.5376	0
cna_cox	cv1	2026	0.5376	0
cna_cox	cv1	3407	0.5376	0
cna_cox	cv2	42	0.4706	0
cna_cox	cv2	2026	0.4706	0
cna_cox	cv2	3407	0.4706	0
cna_cox	cv3	42	0.5559	0
cna_cox	cv3	2026	0.5559	0
cna_cox	cv3	3407	0.5559	0
cna_cox	cv4	42	0.5378	0
cna_cox	cv4	2026	0.5378	0
cna_cox	cv4	3407	0.5378	0
concat_cox	cv0	42	0.5934	0
concat_cox	cv0	2026	0.5934	0
concat_cox	cv0	3407	0.5934	0
concat_cox	cv1	42	0.5787	0
concat_cox	cv1	2026	0.5787	0
concat_cox	cv1	3407	0.5787	0
concat_cox	cv2	42	0.5521	0
concat_cox	cv2	2026	0.5521	0
concat_cox	cv2	3407	0.5521	0
concat_cox	cv3	42	0.6682	0
concat_cox	cv3	2026	0.6682	0
concat_cox	cv3	3407	0.6682	0
concat_cox	cv4	42	0.6252	0
concat_cox	cv4	2026	0.6252	0
concat_cox	cv4	3407	0.6252	0
deepsurv	cv0	42	0.5848	122
deepsurv	cv0	2026	0.5902	56
deepsurv	cv0	3407	0.5931	85
deepsurv	cv1	42	0.5337	4
deepsurv	cv1	2026	0.5599	14
deepsurv	cv1	3407	0.5965	28
deepsurv	cv2	42	0.6031	16

deepsurv	cv2	2026	0.5887	16
deepsurv	cv2	3407	0.6082	10
deepsurv	cv3	42	0.5974	9
deepsurv	cv3	2026	0.6403	11
deepsurv	cv3	3407	0.6225	17
deepsurv	cv4	42	0.6521	49
deepsurv	cv4	2026	0.6684	20
deepsurv	cv4	3407	0.6668	44
gcca_cox	cv0	42	0.5990	0
gcca_cox	cv0	2026	0.5990	0
gcca_cox	cv0	3407	0.5990	0
gcca_cox	cv1	42	0.5709	0
gcca_cox	cv1	2026	0.5709	0
gcca_cox	cv1	3407	0.5709	0
gcca_cox	cv2	42	0.5850	0
gcca_cox	cv2	2026	0.5850	0
gcca_cox	cv2	3407	0.5850	0
gcca_cox	cv3	42	0.6100	0
gcca_cox	cv3	2026	0.6100	0
gcca_cox	cv3	3407	0.6100	0
gcca_cox	cv4	42	0.6267	0
gcca_cox	cv4	2026	0.6267	0
gcca_cox	cv4	3407	0.6267	0
integrao_cox	cv0	42	0.5569	300
integrao_cox	cv0	2026	0.5952	300
integrao_cox	cv0	3407	0.5983	300
integrao_cox	cv1	42	0.5373	300
integrao_cox	cv1	2026	0.5631	300
integrao_cox	cv1	3407	0.5483	300
integrao_cox	cv2	42	0.5825	300
integrao_cox	cv2	2026	0.5781	300
integrao_cox	cv2	3407	0.5737	300
integrao_cox	cv3	42	0.5899	300
integrao_cox	cv3	2026	0.5859	300
integrao_cox	cv3	3407	0.5656	300
integrao_cox	cv4	42	0.5942	300
integrao_cox	cv4	2026	0.5849	300
integrao_cox	cv4	3407	0.5834	300
methylation_cox	cv0	42	0.5534	0
methylation_cox	cv0	2026	0.5534	0
methylation_cox	cv0	3407	0.5534	0
methylation_cox	cv1	42	0.5358	0
methylation_cox	cv1	2026	0.5358	0
methylation_cox	cv1	3407	0.5358	0
methylation_cox	cv2	42	0.5254	0
methylation_cox	cv2	2026	0.5254	0
methylation_cox	cv2	3407	0.5254	0
methylation_cox	cv3	42	0.5771	0
methylation_cox	cv3	2026	0.5771	0
methylation_cox	cv3	3407	0.5771	0
methylation_cox	cv4	42	0.5294	0
methylation_cox	cv4	2026	0.5294	0
methylation_cox	cv4	3407	0.5294	0
mogonet_cox	cv0	42	0.6117	7
mogonet_cox	cv0	2026	0.6165	15
mogonet_cox	cv0	3407	0.6115	8

mogonet_cox	cv1	42	0.5765	25
mogonet_cox	cv1	2026	0.5864	8
mogonet_cox	cv1	3407	0.5716	19
mogonet_cox	cv2	42	0.5789	20
mogonet_cox	cv2	2026	0.5922	20
mogonet_cox	cv2	3407	0.5836	20
mogonet_cox	cv3	42	0.6334	5
mogonet_cox	cv3	2026	0.6504	5
mogonet_cox	cv3	3407	0.6317	4
mogonet_cox	cv4	42	0.5910	132
mogonet_cox	cv4	2026	0.6183	4
mogonet_cox	cv4	3407	0.6199	49
mrna_cox	cv0	42	0.5999	0
mrna_cox	cv0	2026	0.5999	0
mrna_cox	cv0	3407	0.5999	0
mrna_cox	cv1	42	0.5742	0
mrna_cox	cv1	2026	0.5742	0
mrna_cox	cv1	3407	0.5742	0
mrna_cox	cv2	42	0.5954	0
mrna_cox	cv2	2026	0.5954	0
mrna_cox	cv2	3407	0.5954	0
mrna_cox	cv3	42	0.6568	0
mrna_cox	cv3	2026	0.6568	0
mrna_cox	cv3	3407	0.6568	0
mrna_cox	cv4	42	0.6695	0
mrna_cox	cv4	2026	0.6695	0
mrna_cox	cv4	3407	0.6695	0
multisurv	cv0	42	0.5978	12
multisurv	cv0	2026	0.6078	21
multisurv	cv0	3407	0.6027	15
multisurv	cv1	42	0.5301	7
multisurv	cv1	2026	0.5401	7
multisurv	cv1	3407	0.5304	7
multisurv	cv2	42	0.6007	67
multisurv	cv2	2026	0.5982	54
multisurv	cv2	3407	0.5968	69
multisurv	cv3	42	0.5689	15
multisurv	cv3	2026	0.5654	16
multisurv	cv3	3407	0.6176	29
multisurv	cv4	42	0.5731	16
multisurv	cv4	2026	0.6083	5
multisurv	cv4	3407	0.6512	6
no_anchor	cv0	42	0.6322	5
no_anchor	cv0	2026	0.6059	3
no_anchor	cv0	3407	0.6055	4
no_anchor	cv1	42	0.5663	7
no_anchor	cv1	2026	0.5653	25
no_anchor	cv1	3407	0.5633	18
no_anchor	cv2	42	0.5751	53
no_anchor	cv2	2026	0.5833	78
no_anchor	cv2	3407	0.6007	74
no_anchor	cv3	42	0.6279	12
no_anchor	cv3	2026	0.6187	8
no_anchor	cv3	3407	0.6396	8
no_anchor	cv4	42	0.5844	3
no_anchor	cv4	2026	0.6167	14

no_anchor	cv4	3407	0.6313	15
no_graph	cv0	42	0.6096	4
no_graph	cv0	2026	0.6197	5
no_graph	cv0	3407	0.6142	2
no_graph	cv1	42	0.5535	6
no_graph	cv1	2026	0.5776	24
no_graph	cv1	3407	0.5719	20
no_graph	cv2	42	0.6071	94
no_graph	cv2	2026	0.5862	129
no_graph	cv2	3407	0.5784	70
no_graph	cv3	42	0.6305	12
no_graph	cv3	2026	0.6234	6
no_graph	cv3	3407	0.6195	18
no_graph	cv4	42	0.5960	3
no_graph	cv4	2026	0.6167	23
no_graph	cv4	3407	0.6195	14
no_mask	cv0	42	0.6111	5
no_mask	cv0	2026	0.5951	4
no_mask	cv0	3407	0.6093	1
no_mask	cv1	42	0.5635	8
no_mask	cv1	2026	0.5655	11
no_mask	cv1	3407	0.5499	15
no_mask	cv2	42	0.5955	95
no_mask	cv2	2026	0.5955	87
no_mask	cv2	3407	0.5692	99
no_mask	cv3	42	0.6265	13
no_mask	cv3	2026	0.6198	8
no_mask	cv3	3407	0.6227	8
no_mask	cv4	42	0.5837	3
no_mask	cv4	2026	0.6351	35
no_mask	cv4	3407	0.6171	14
no_residual	cv0	42	0.6092	6
no_residual	cv0	2026	0.6075	13
no_residual	cv0	3407	0.6113	2
no_residual	cv1	42	0.5771	8
no_residual	cv1	2026	0.5694	43
no_residual	cv1	3407	0.5718	3
no_residual	cv2	42	0.5998	138
no_residual	cv2	2026	0.6162	75
no_residual	cv2	3407	0.6260	19
no_residual	cv3	42	0.6170	35
no_residual	cv3	2026	0.6324	6
no_residual	cv3	3407	0.6286	7
no_residual	cv4	42	0.6398	33
no_residual	cv4	2026	0.6110	9
no_residual	cv4	3407	0.6159	14
original_graph	cv0	42	0.5977	1
original_graph	cv0	2026	0.5960	1
original_graph	cv0	3407	0.5385	45
original_graph	cv1	42	0.5759	15
original_graph	cv1	2026	0.5881	12
original_graph	cv1	3407	0.5754	12
original_graph	cv2	42	0.5670	19
original_graph	cv2	2026	0.5864	291
original_graph	cv2	3407	0.5915	51
original_graph	cv3	42	0.6163	135

original_graph	cv3	2026	0.6162	113
original_graph	cv3	3407	0.6140	63
original_graph	cv4	42	0.6168	10
original_graph	cv4	2026	0.5995	7
original_graph	cv4	3407	0.5915	8
pairwise_cox	cv0	42	0.6316	0
pairwise_cox	cv0	2026	0.6316	0
pairwise_cox	cv0	3407	0.6316	0
pairwise_cox	cv1	42	0.5965	0
pairwise_cox	cv1	2026	0.5965	0
pairwise_cox	cv1	3407	0.5965	0
pairwise_cox	cv2	42	0.6042	0
pairwise_cox	cv2	2026	0.6042	0
pairwise_cox	cv2	3407	0.6042	0
pairwise_cox	cv3	42	0.6416	0
pairwise_cox	cv3	2026	0.6416	0
pairwise_cox	cv3	3407	0.6416	0
pairwise_cox	cv4	42	0.6214	0
pairwise_cox	cv4	2026	0.6214	0
pairwise_cox	cv4	3407	0.6214	0
proposed	cv0	42	0.6143	5
proposed	cv0	2026	0.6089	4
proposed	cv0	3407	0.6118	2
proposed	cv1	42	0.5663	7
proposed	cv1	2026	0.5653	25
proposed	cv1	3407	0.5633	18
proposed	cv2	42	0.6082	70
proposed	cv2	2026	0.6069	155
proposed	cv2	3407	0.6264	29
proposed	cv3	42	0.6347	24
proposed	cv3	2026	0.6307	7
proposed	cv3	3407	0.6301	8
proposed	cv4	42	0.5844	3
proposed	cv4	2026	0.6167	14
proposed	cv4	3407	0.6313	15
rsf	cv0	42	0.6016	0
rsf	cv0	2026	0.6040	0
rsf	cv0	3407	0.5955	0
rsf	cv1	42	0.5620	0
rsf	cv1	2026	0.5601	0
rsf	cv1	3407	0.5715	0
rsf	cv2	42	0.6257	0
rsf	cv2	2026	0.6190	0
rsf	cv2	3407	0.6151	0
rsf	cv3	42	0.6529	0
rsf	cv3	2026	0.6480	0
rsf	cv3	3407	0.6524	0
rsf	cv4	42	0.6559	0
rsf	cv4	2026	0.6727	0
rsf	cv4	3407	0.6568	0
structured_clinical	cv0	42	0.6563	6
structured_clinical	cv0	2026	0.6545	38
structured_clinical	cv0	3407	0.6551	3
structured_clinical	cv1	42	0.6079	16
structured_clinical	cv1	2026	0.6022	44
structured_clinical	cv1	3407	0.5764	6

structured_clinical	cv2	42	0.6438	81
structured_clinical	cv2	2026	0.6487	14
structured_clinical	cv2	3407	0.6463	22
structured_clinical	cv3	42	0.6943	10
structured_clinical	cv3	2026	0.6928	7
structured_clinical	cv3	3407	0.6939	28
structured_clinical	cv4	42	0.6904	28
structured_clinical	cv4	2026	0.6661	11
structured_clinical	cv4	3407	0.6921	22

9 Training-only canonical interpretation

The following examples are the five largest absolute projection coefficients for component one of fold zero, separately per molecular view. Structure correlations are Pearson correlations of each standardized input feature with that training canonical score. The complete fold/component table is in canonical_loadings.csv. Neither coefficients nor correlations establish causal biomarkers; signs and rotations are not compared across folds.

View	Gene/probe row	Coefficient	Correlation
mRNA	PGAP3|row5492	-0.133	-0.644
mRNA	ERBB2|row11007	-0.125	-0.642
mRNA	GRB7|row15911	-0.123	-0.766
mRNA	GSDMB|row16946	-0.088	-0.697
mRNA	NUPR2|row611	0.038	0.049
Methylation	MT1M|row7362	-0.126	-0.352
Methylation	MED24|row6660	-0.109	-0.146
Methylation	SBNO2|row9938	-0.105	-0.215
Methylation	PREX2|row8931	-0.100	-0.129
Methylation	ZNF516|row12959	-0.088	-0.239
CNA	PGAP3|row14718	-0.089	-0.823
CNA	MIR4728|row11734	-0.088	-0.822
CNA	ERBB2|row5435	-0.088	-0.822
CNA	MIEN1|row10844	-0.085	-0.821
CNA	STARD3|row19045	-0.083	-0.823

Source data, clinical templates and patient-level outcomes/predictions remain in the ignored local private directory. The public release contains code, configurations, source hashes and aggregate tables only. Author identity, institution, funding, conflicts and ethics review/waiver determination remain for responsible-author completion. AI assistance is disclosed in the main draft. Neither this appendix nor its page count is assumed compatible with the conference's technical-page submission restrictions.